

Cities Environmental and Climate Grids Data Dictionary			
Attribute ID (REVISED)	Unit	Time	Attribute Description
ID_UC_G0	Urban Center	Current	Unique city ID
gridcode	Gridcell	Current	Unique ID of grid cell (starts at "1" in each city)
ID_UC_gridcode	Gridcell	Current	Combination of ID_UC_G0 and gridcode to make a unique ID for every grid cell in the corresponding city
UC_CNT_NM	Urban Center	Current	Country name
UC_UCN_NM	Urban Center	Current	Main name associated with the urban center
UC_UCA_KM2	Urban Center	Current	Area of the urban center in km2
UC_DEV_WIG	Urban Center	Current	World Bank Income Group associated with the country
UC_DEV_USR	Urban Center	Current	UN SDG Region associated with the country
UC_GDP20	Urban Center	Current	Average of 2020 total GDP per 30-arcsecond sq., measured in PPP, 2017 international dollars.
POP20_GRID	Gridcell	Current	Total population of the grid in 2020
POP20_UC	Urban Center	Current	Total population of the urban center in 2020
POP20_0_14	Urban Center	Current	Share of young population (0-14 years)
POP20_15_64	Urban Center	Current	Share of adult population (15-64 years)
POP20_65UP	Urban Center	Current	Share of old population (>=65 years)
POP20_DEP_RATIO	Urban Center	Current	Dependency Ratio as calculated as population share (age 0-14 + age 65 and older) / (age 15-64)
POP25_GRID	Gridcell	Current	Total population of the grid in 2025
POP25_LOWSES_GRID	Gridcell	Current	Total population of the grid in 2025 living in low socioeconomic status areas ("SES_2CAT_P" = 1)
POP25_UC	Urban Center	Current	Total population of the urban center in 2025
POP25_UC_CAT	Urban Center	Current	Total city population in 2025, categorized: 1: <50k, 2: 50k-100k, 3: 100k-300k, 4: 300k+
POP25_UC_U300	Urban Center	Current	Flag for "small and medium sized" cities, defined as 2025 population < 300,000
L_DIST_1M	Urban Center	Current	Distance (km) to nearest city of 1M+ inhabitants generated from GHS-POP R2023A
L_DIST_1M_NM	Urban Center	Current	Name of nearest city of 1M+ inhabitants taken from GHS-UCDB R2019A v1.2
L_DIST_500K	Urban Center	Current	Distance (km) to nearest city of 500k+ inhabitants generated from GHS-POP R2023A
L_DIST_500K_NM	Urban Center	Current	Name of nearest city of 500k+ inhabitants taken from GHS-UCDB R2019A v1.2
L_ELEV_GRID	Gridcell	Current	Average elevation of the grid cell (in meters above sealevel)
L_ELEV_UC	Urban Center	Current	Average elevation of the city (in meters above sealevel)
L_LECZ_GRID	Gridcell	Current	Flag for gridcells located within a 10km low elevation coastal zone
L_LECZ_UC	Urban Center	Current	Flag for urban centers fully or partially located in 10km low elevation coastal zone
L_DRY_GRID	Gridcell	Current	Flag if the climate zone in the grid cell is equal to any of the following: BWh Arid, desert, hot; BWk Arid, desert, cold; BSh Arid, steppe, hot; BSk Arid, steppe, cold; ET Polar, tundra; EF Polar, frost. (Zones: 4, 5, 6, 7, 29, 30, respectively)
L_DRY_UC	Urban Center	Current	A flag if any grid cell in the urban center is located in one of the following climate zones: BWh Arid, desert, hot; BWk Arid, desert, cold; BSh Arid, steppe, hot; BSk Arid, steppe, cold; ET Polar, tundra; EF Polar, frost. (Zones: 4, 5, 6, 7, 29, 30, respectively)
VULN_TEMP_VAL	Urban Center	Current	Number of days in 2016 that maximum (dry) air temperature exceeded 45C (113F) in the urban center
VULN_TEMP	Urban Center	Current	Flag that the urban center was exposed to 7+ days of (dry) air temperature exceeding 45C (113F) in 2016
VULN_WBGT_VAL	Urban Center	Current	Number of days in 2016 that Wet Bulb Globe Temperature (humid heat) exceeded 30C (86F) in the urban center
VULN_WBGT	Urban Center	Current	Flag that the urban center was exposed to 7+ days of (humid) Wet Bulb Globe Temperature exceeding 30C (86F) in 2016
VULN_WSDI_VAL	Urban Center	Current	The annual average (2010-2021) number of consecutive days that temperature exceeding 90% the historical average (1880-2021) in the urban center
VULN_WSDI	Urban Center	Current	Flag that the urban center was recently (annual average 2010-2021) exposed to 7+ consecutive days of temperature exceeding 90% the historical average (1880-2021)
VULN_HEAT_VAL	Urban Center	Current	Number of days in 2016 that the urban center experienced extreme heat (maximum of VULN_TEMP_VAL, VULN_WBGT_VAL)
VULN_HEAT	Urban Center	Current	Flag that the urban center was exposed to 7+days of extreme heat in 2016 (TEMP_GRID or WBGT_GRID ==1)
VULN_WATR_VAL	Urban Center	Current	"Worst" water scarcity classification among all watersheds within the urban center in 2016, where 0: Non-water scarcity; 3: Arid and low water use; 4: Seasonal water scarcity; 5: Perennial water scarcity
VULN_WATR	Urban Center	Current	Flag that population of gridcell was exposed to any level of water scarcity in 2016 (arid conditions, or seasonal or perennial water scarcity)
VULN_CYCL_VAL	Urban Center	Current	Maximum IBTrACS cyclone wind speed in km/hr in the city from 1991-01-28 — 2024-04-08
VULN_CYCL	Gridcell	Current	Flag that population of gridcell experienced Category 1 hurricane cyclone winds between 1991-01-28 — 2024-04-08 (max wind speed >=119km/hr or 74 mph)
VULN_SURG_VAL	Gridcell	Current	Percent of the population exposed to storm surge with a 10-year return period normalized with dem > 0 as a share of the total population in the grid cell.
VULN_SURG	Gridcell	Current	Flag that a majority of the population of gridcell likely experienced a storm surge flood event by 2025 (as proxied by a 1 in 10 year storm surge flood event)
VULN_RIVR_VAL	Gridcell	Current	Percent of the population exposed to 1+ months of seasonal riverine flooding in "Aggregated Seasonality" (as measured in ~100x100m grids)
VULN_RIVR	Gridcell	Current	Flag that a majority of the population of the gridcell are exposed to seasonal riverine flooding
VULN_AIR_VAL	Gridcell	Current	Average PM2.5 concentrations between 2013-2022

VULN_AIR_CNT	Gridcell	Current	Count of years where PM2.5 >10 mcg 2013-2022
VULN_AIR	Gridcell	Current	Flag that population of gridcell was exposed to average annual PM2.5 pollution exceeding 10 µg/m3 in 5 or more of the past 10 years
RISK_TEMP_VAL	Urban Center	Future	Number of days that maximum (dry) air temperature (under SSP245) is expected to exceed 45C (113F) in 2050
RISK_TEMP	Urban Center	Future	Flag that the urban center is expected to be exposed to 7+ days where maximum (dry) air temperature exceeds 45C (113F) in 2050 (under SSP245)
RISK_WBGT_VAL	Urban Center	Future	Number of days in 2050 that Web Bulb Globe Temperature (under SSP245) is expected to exceed 30C (86F)
RISK_WBGT	Urban Center	Future	Flag that the urban center is expected to be exposed to 7+ days where Web Bulb Globe Temperature exceeds 30C (86F) in 2050 (under SSP245)
RISK_HEAT_VAL	Urban Center	Future	Number of days in 2050 that urban center is expected to be exposed to extreme heat (maximum of RISK_TEMP_VAL, RISK_WBGT_VAL)
RISK_HEAT	Urban Center	Future	Flag that the urban center is expected to experience 7+days/year of extreme heat in 2050 (under SSP245) (RISK_TEMP or RISK_WBGT ==1)
RISK_WATR_VAL	Urban Center	Future	"Worst" water scarcity classification among all watersheds within the urban center in 2050 (under SSP245) where 0: No water scarcity; 3: Arid & low water use; 4: Seasonal water scarcity; 5: Perennial water scarcity
RISK_WATR	Urban Center	Future	Flag that poulation of gridcell is expected to be exposed to any level of water scarcity in 2050 (under SSP245), meaning arid conditions, or seasonal or perennial water scarcity
RISK_CYCL_VAL	Urban Center	Future	Maximum storm wind speed expected in the city by 2050 (as proxied by a 1 in 100 year event)
RISK_CYCL_CAT	Urban Center	Future	Maximum storm wind speed expected in the city by 2050 (as proxied by a 1 in 100 year event), classified by Saffir-Simpson Hurricane Wind Scale (National Weather Service) of hurricanes: 0: <119 km/hr 1: 119-153 km/hr, 2: 154-177 km/hr, 3: 178-208 km/hr, 4: 209-251 km/hr 5: >251 km/hr
RISK_CYCL	Gridcell	Future	Flag that population of gridcell is expected to experience Cagtegrory 1 hurricane force winds by 2050 (as proxied by a 1 in 100 year max wind speed >=119km/hr (74 mph)
SURG_RISK_COMPARE	Urban Center	Future	Percent of 2025 population in LECZ expected to experience a 1 in 100 year surge event
RISK_SURG_VAL	Gridcell	Future	Percent of the grid cell population in LECZ exposed to storm surge with a 100-year return period
RISK_SURG	Gridcell	Future	Flag that a majority of the gridcell population is expected to experience a storm surge flood event by 2050 (as proxied by a 1 in 100 year storm surge flood event)
RISK_RIVR_VAL	Gridcell	Future	Percent of the current population expected to experience seasonal riverine flooding by 2050 (as proxied by 1 in 100 yr riverine flooding in ~1x1km grids and GHS-POP 2025)
RISK_RIVR	Gridcell	Future	Flag that a majority of the population of the gridcell is expected to experience seasonal riverine flooding by 2050
RISK_LAND_VAL	Gridcell	Future	Level of landslide susceptibility where 0=no susceptibility and 5=very high susceptibility that takes up majority of the area of a given cell
RISK_LAND_CAT	Urban Center	Future	Average level of landslide susceptibility where 0=no susceptibility and 5=very high susceptibility
RISK_LAND	Gridcell	Future	Flag that landslide susceptibility is high (4) or very high (5)